

AP CALCULUS AB – UNIT 1

Limits and Continuity

Expanded Complete-Coverage Coloured Review



Name: _____ Class: _____ Date: _____

Review Scope and Learning Objectives

- Interpret and estimate limits from symbolic, graphical, and numerical representations.
- Evaluate limits using limit laws, algebraic manipulation, and the Squeeze Theorem.
- Analyze continuity, classify discontinuities, and repair removable discontinuities.
- Connect infinite and end-behavior limits with asymptotes and apply the Intermediate Value Theorem.
- Review every idea, theorem hypothesis, representation, procedure, and common misconception appearing in the website lessons.

Complete Lesson Coverage Audit

Every website lesson is given its own mastery section below. Each section contains the complete lesson objectives, strategic reminders, a solved model, and foundation-to-challenge practice. The final synthesis sections mix lessons so that students must select procedures rather than follow a stated template.

Complete Website Lesson Coverage Map

Lesson	Website title	Ideas explicitly reviewed
1.1	Can Change Occur at an Instant?	Interpret average rate of change in a context, including correct units.; Use shrinking intervals to estimate a rate at one instant.; Connect secant slopes, tangent slopes, and limit notation.; Explain why an instantaneous rate is defined through nearby values rather than a zero-length interval.
1.2	Defining Limits and Using Limit Notation	Interpret every symbol in finite, one-sided, and infinite limit notation.; Translate among words, graphs, tables, and limit statements.; Use one-sided agreement to determine whether a two-sided limit exists.; Distinguish a limit, a point value, an infinite limit, and a statement that the limit does not exist.
1.3	Estimating Limit Values from Graphs	Trace a graph toward a target from each side before reading any point marker.; Estimate one-sided and two-sided limits from graphical evidence.; Distinguish holes, jumps, vertical asymptotes, endpoints, corners, and oscillation.; State the precision supported by the scale and viewing window.
1.4	Estimating Limit Values from Tables	Select input values that approach the target from both sides.; Estimate finite, infinite, and nonexistent limits from tabular patterns.; Distinguish the center value from nearby evidence.; Evaluate the reliability of a table, including rounding, spacing, and deceptive sampling.
1.5	Determining Limits Using Algebraic Properties of Limits	Apply constant, sum, difference, product, quotient, power, root, and composition laws.; Check every hypothesis before using a limit law.; Separate known limit values from function values at the target.; Justify direct substitution through continuity and domain conditions.
1.6	Determining Limits Using Algebraic Manipulation	Use substitution as a diagnostic rather than a final answer when it produces an indeterminate form.; Select factoring, conjugates, common denominators, trigonometric identities, or side-specific algebra efficiently.; Justify cancellation through equality on a punctured neighborhood.; Use the standard trigonometric limits in radians.

1.7	Selecting Procedures for Determining Limits	Diagnose a limit before calculating.; Match algebraic structure to an efficient valid procedure.; Choose among analytical, graphical, tabular, one-sided, and theorem-based evidence.; Verify a result through a second representation or local check.
1.8	Determining Limits Using the Squeeze Theorem	State all hypotheses of the Squeeze Theorem.; Create useful upper and lower bounds from absolute values and trigonometric inequalities.; Distinguish persistent oscillation from shrinking oscillation.; Write a complete AP-style squeeze justification.
1.9	Connecting Multiple Representations of Limits	Extract one-sided limits, two-sided limits, and point values from any representation.; Translate accurately among formula, graph, table, words, and notation.; Distinguish exact conclusions from numerical estimates.; Audit whether multiple representations are locally coherent.
1.10	Exploring Types of Discontinuities	Classify removable, jump, and infinite discontinuities using one-sided evidence.; Distinguish the three named types from oscillatory or domain-gap failures.; Use factor cancellation and multiplicity to classify rational-function discontinuities.; Write the limit evidence before naming the type.
1.11	Defining Continuity at a Point	State and apply the three conditions for continuity at a point.; Identify the first failed condition from graphs, tables, formulas, and piecewise rules.; Use continuity of standard functions and valid compositions.; Reject common invalid converses about limits and point values.
1.12	Confirming Continuity over an Interval	Distinguish continuity on open, closed, and half-open intervals.; Use one-sided continuity at included endpoints.; Find maximal intervals of continuity from domain restrictions.; Justify continuity of sums, products, quotients, roots, and compositions on stated intervals.
1.13	Removing Discontinuities	Decide whether a discontinuity can be removed by changing a point value.; Find parameters that make piecewise functions continuous at joins.; Create or cancel common factors to identify removable holes.; Solve multi-boundary continuity systems and justify uniqueness or impossibility.
1.14	Connecting Infinite Limits and Vertical Asymptotes	Interpret positive and negative infinite limits as directional unbounded behavior.; Determine one-sided signs near denominator zeros.; Use factor multiplicity and cancellation to distinguish holes from vertical asymptotes.; Connect formulas, sign charts, tables, graphs, and asymptote notation.
1.15	Connecting Limits at Infinity and Horizontal Asymptotes	Evaluate limits as $x \rightarrow \infty$ and $x \rightarrow -\infty$.; Determine horizontal asymptotes from end limits.; Use degree comparison, dominant powers, radicals, conjugates, and growth hierarchy.; Analyze the two ends independently and handle $\sqrt{x^2} = x $ correctly.
1.16	Working with the Intermediate Value Theorem	State every hypothesis of the Intermediate Value Theorem.; Use endpoint values to guarantee roots or other target outputs.; Count minimum distinct solutions from disjoint sign-changing intervals.; Use auxiliary functions and bisection to localize guaranteed solutions.

Unit Knowledge, Definitions, and Theorem Library

Key Knowledge, Formulas, and Theorems

- Average rate on $[a, b]$: $\frac{f(b) - f(a)}{b - a}$; an instantaneous rate is the limit of such secant slopes as the interval shrinks.
- Finite limit notation separates nearby behavior from the point value. A two-sided limit exists exactly when the two one-sided limits agree.
- Precise finite-limit definition: $\lim_{x \rightarrow c} f(x) = L$ means $\forall \varepsilon > 0 \exists \delta > 0$ such that $0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon$.
- The negation of the precise definition requires one fixed $\varepsilon_0 > 0$ for which every $\delta > 0$ admits a violating

input.

- Limit laws require the component limits to exist; the quotient law also requires a nonzero limiting denominator.
- Direct substitution is justified for a function continuous at the target and within its real domain.
- Indeterminate forms such as $0/0$ signal the need for a rewrite; they are not answers. Common tools are factoring, conjugates, common denominators, identities, and side-specific simplification.
- Standard trigonometric limits are used in radians: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ and $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$.
- Squeeze Theorem: if $g(x) \leq f(x) \leq h(x)$ near c and both outer limits equal L , then $\lim_{x \rightarrow c} f(x) = L$.
- For compositions, track the landing value and its direction before applying the outer function, especially for one-sided and piecewise data.
- Continuity at c requires: $f(c)$ exists, $\lim_{x \rightarrow c} f(x)$ exists, and the two values agree. Endpoint continuity uses the included one-sided limit.
- Discontinuities may be removable, jump, infinite, oscillatory, or domain-gap failures. Only a finite common limit can be repaired by redefining one point value.
- Factor multiplicity and sign charts determine one-sided infinite limits and vertical-asymptote branch directions.
- End limits are analyzed separately. For radicals, $\sqrt{x^2} = |x|$; for rational functions, compare degrees and leading coefficients.
- IVT requires continuity on $[a, b]$ and guarantees at least one input attaining every output between $f(a)$ and $f(b)$; it gives neither uniqueness nor an exact location.

Precise Limit Definition and Proof Pattern

$$\lim_{x \rightarrow c} f(x) = L \iff \forall \varepsilon > 0 \exists \delta > 0 : 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

A direct proof begins with an arbitrary $\varepsilon > 0$, works backward to design a sufficient δ , then works forward from $0 < |x - c| < \delta$ to the desired output inequality. The point $x = c$ is excluded because a limit describes nearby behavior.

Solved Example: Precise-definition model: $\lim_{x \rightarrow 2} x^2 = 4$

Choose $\delta = \min\{1, \varepsilon/5\}$. If $0 < |x - 2| < \delta$, then $|x + 2| < 5$ and

$$|x^2 - 4| = |x - 2||x + 2| < 5\delta \leq \varepsilon.$$

Thus the precise definition is satisfied.

Strategy and Precision Checklist

1. Name the representation and the requested quantity before calculating.
2. State all hypotheses before using a theorem.
3. Preserve exact values in non-calculator work and use radians for trigonometric calculus.
4. A graph or table supports an estimate; an analytic proof requires a definition, theorem, or valid algebraic argument.
5. Contextual conclusions must include the instant, sign, units, and meaning.
6. Candidate conditions such as a zero derivative, zero second derivative, or zero denominator require sign/domain verification before a conclusion.

Lesson 1.1: Can Change Occur at an Instant?**Complete objectives**

- Interpret average rate of change in a context, including correct units.
- Use shrinking intervals to estimate a rate at one instant.
- Connect secant slopes, tangent slopes, and limit notation.
- Explain why an instantaneous rate is defined through nearby values rather than a zero-length interval.

Strategic reminders

1. Compute an average rate first. Then shorten the interval while keeping the target instant fixed.
2. For a left estimate use inputs below the target; for a right estimate use inputs above the target. Agreement supports a two-sided instantaneous rate.
3. A numerical estimate should include units and a contextual sentence. A symbolic answer should display the difference quotient before simplification.

Solved Example: Polynomial motion

A particle has position $s(t) = t^3 - 2t$ meters. Estimate its instantaneous velocity at $t = 2$ using $[2, 2.01]$ and $[1.99, 2]$.

Answer and Reasoning

$s(2) = 4$. The right secant slope is $\frac{s(2.01) - s(2)}{0.01} = 10.0601$ and the left secant slope is $\frac{s(2) - s(1.99)}{0.01} = 9.9401$. Both approach 10, so the instantaneous velocity is approximately 10 m/s. In exact limit form, $\lim_{h \rightarrow 0} \frac{(2+h)^3 - 2(2+h) - 4}{h} = 10$.

Lesson Mastery Practice

1. For $s(t) = t^2 + 3t$, find the average rate of change on $[1, 4]$.

Foundation

2. For $s(t) = t^3 - 2t$, calculate the secant slope on $[2, 2.1]$.

Core

3. A fish population is modeled by $P(t) = 1000 + 40t - 3t^2$. Find the instantaneous rate of change at $t = 4$ using a limit or an equivalent algebraic derivation.

Advanced

4. Construct a quadratic position function $s(t) = at^2 + bt + c$ satisfying $s(0) = 2$, $s(1) = 5$, and instantaneous velocity 7 at $t = 1$.

Challenge

AP-Style Reasoning Check

For $f(x) = x^2 - 4x$, which expression represents the instantaneous rate at $x = 3$?

(A) $\frac{f(3)-f(3)}{3-3}$

(B) $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$

(C) $\lim_{x \rightarrow 0} \frac{f(x) - f(3)}{x - 3}$

(D) $\frac{f(4)-f(2)}{2}$

Lesson 1.2: Defining Limits and Using Limit Notation**Complete objectives**

- Interpret every symbol in finite, one-sided, and infinite limit notation.
- Translate among words, graphs, tables, and limit statements.
- Use one-sided agreement to determine whether a two-sided limit exists.
- Distinguish a limit, a point value, an infinite limit, and a statement that the limit does not exist.

Strategic reminders

1. Read a limit statement in the order: expression, input target, input direction, output destination.
2. Always separate $\lim_{x \rightarrow c} f(x)$ from $f(c)$. Either may exist without the other.
3. Before writing a two-sided limit, record the left- and right-hand limits. If they differ, the two-sided limit does not exist.

Solved Example: Words to symbols

Write: “As x approaches -3 from values greater than -3 , $p(x)$ decreases without bound.”

Answer and Reasoning

$$\lim_{x \rightarrow -3^+} p(x) = -\infty.$$

Lesson Mastery Practice

5. Translate $\lim_{x \rightarrow 5} f(x) = 9$ into a complete sentence.

Foundation

6. Given $\lim_{x \rightarrow 3^-} f(x) = 8$ and $\lim_{x \rightarrow 3^+} f(x) = 8$, state the two-sided limit.

Core

7. Suppose $\lim_{x \rightarrow a} f(x) = L$. Must $f(a)$ exist? Must L be finite? Answer for the stated notation.

Advanced

Challenge

If $\lim_{x \rightarrow 2^-} f(x) = 5$ and $\lim_{x \rightarrow 2^+} f(x) = 5$, which must be true?

Lesson 1.3: Estimating Limit Values from Graphs**Complete objectives**

- Trace a graph toward a target from each side before reading any point marker.
- Estimate one-sided and two-sided limits from graphical evidence.
- Distinguish holes, jumps, vertical asymptotes, endpoints, corners, and oscillation.
- State the precision supported by the scale and viewing window.

Strategic reminders

1. Cover the filled point with your finger and trace the curve from the left and right.
2. Write the two one-sided limits before deciding the two-sided limit.
3. For a vertical asymptote, inspect branch direction rather than merely naming the vertical line.

Solved Example: Hole with a different point value

In Graph A, determine $\lim_{x \rightarrow 1} f(x)$ and $f(1)$.

Answer and Reasoning

The curve approaches the open point $(1, 1)$ from both sides, so the limit is 1. The filled point is at $(1, 4)$, so $f(1) = 4$.

Lesson Mastery Practice

9. Using Graph A, find $\lim_{x \rightarrow 1^-} f(x)$.

Foundation

10. Using Graph A, explain why changing the filled point would not change the limit at $x = 1$.

Core

11. A graphing window shows a curve approaching about 2.4 near $x = 5$, with vertical grid spacing 1. Which is more defensible: 2.4 or 2.41763? Explain.

Advanced

12. Draw a graph with $f(0) = 2$, $\lim_{x \rightarrow 0} f(x) = -1$, and a vertical asymptote at $x = 3$ with both sides tending to $+\infty$. Label all special points.

Challenge

AP-Style Reasoning Check

At $x = a$, a graph approaches 6 from both sides but has a filled point at height -3 . Which statement is true?

(A) $\lim_{x \rightarrow a} f(x) = -3$

(B) $\lim_{x \rightarrow a} f(x) = 6$

(C) The limit DNE.

(D) $f(a) = 6$

Lesson 1.4: Estimating Limit Values from Tables**Complete objectives**

- Select input values that approach the target from both sides.
- Estimate finite, infinite, and nonexistent limits from tabular patterns.
- Distinguish the center value from nearby evidence.
- Evaluate the reliability of a table, including rounding, spacing, and deceptive sampling.

Strategic reminders

1. Build two columns of inputs: $c - 10^{-k}$ and $c + 10^{-k}$.
2. Compare both sides separately before forming a two-sided conclusion.
3. Use calculator precision high enough to avoid rounding a hole, jump, or oscillation into a false pattern.

Solved Example: Finite estimate

Estimate the limit from the table.

x	3.9	3.99	3.999	4.001	4.01	4.1
$f(x)$	7.81	7.9801	7.998001	8.002001	8.0201	8.21

Answer and Reasoning

Both sides approach 8. The stable digits support $\lim_{x \rightarrow 4} f(x) = 8$.

Lesson Mastery Practice

13. Use the first shared table to estimate $\lim_{x \rightarrow 4^-} f(x)$.

Foundation

14. Create four suitable x -values to estimate a two-sided limit at $x = 7$.

Core

15. A calculator rounds all outputs to two decimals and reports 1.00 on both sides. Explain why this does not prove the limit is exactly 1.

Advanced

16. Give two functions that agree at every input in the finite table $x = \pm 0.1, \pm 0.01, \pm 0.001$ but have different limits at 0.

Challenge

AP-Style Reasoning Check

A table near $x = 5$ has outputs approaching -2 from both sides, while the center row is 17. Which conclusion is best?

(A) $\lim_{x \rightarrow 5} f(x) = 17$

(B) $\lim_{x \rightarrow 5} f(x) = -2$

(C) The limit DNE.

(D) No conclusion is possible.

Lesson 1.5: Determining Limits Using Algebraic Properties of Limits**Complete objectives**

- Apply constant, sum, difference, product, quotient, power, root, and composition laws.
- Check every hypothesis before using a limit law.
- Separate known limit values from function values at the target.
- Justify direct substitution through continuity and domain conditions.

Strategic reminders

1. Write the limit of each changing component before combining them.
2. A failed hypothesis means a law is inconclusive, not that the limit automatically fails.
3. Direct substitution is a theorem-based shortcut for continuous functions at allowed inputs.

Solved Example: Linear combination

Given $\lim_{x \rightarrow a} f(x) = 5$ and $\lim_{x \rightarrow a} g(x) = -2$, find $\lim_{x \rightarrow a} [3f(x) - 4g(x) + 1]$.

Answer and Reasoning

$$3(5) - 4(-2) + 1 = 24.$$

Lesson Mastery Practice

17. Using the shared limit data, find $\lim_{x \rightarrow 2} [f(x) + g(x)]$.

Foundation

18. Using the shared data, evaluate $\lim_{x \rightarrow 2} \sqrt{f(x) + 5}$.

Core

19. If $f(x) \rightarrow 4$ and $g(x) \rightarrow 0$, can the quotient law evaluate $\lim f(x)/g(x)$? Explain precisely.

Advanced

20. Let $f(x) \rightarrow a$ and $g(x) \rightarrow b$. Find a condition on a, b that guarantees $\lim \frac{f(x)^2 - g(x)^2}{f(x) - g(x)} = a + b$ using only limit laws.

Challenge

AP-Style Reasoning Check

If $\lim f = 2$ and $\lim g = -5$, then $\lim(3f - g^2)$ equals

(A) -19

(B) -7

(C) 19

(D) 31

Lesson 1.6: Determining Limits Using Algebraic Manipulation**Complete objectives**

- Use substitution as a diagnostic rather than a final answer when it produces an indeterminate form.
- Select factoring, conjugates, common denominators, trigonometric identities, or side-specific algebra efficiently.
- Justify cancellation through equality on a punctured neighborhood.
- Use the standard trigonometric limits in radians.

Strategic reminders

1. First substitute and classify the result. Then choose the shortest structure-preserving rewrite.
2. State the excluded target when canceling a factor; the simplified expression is used only for nearby inputs.
3. For absolute value and piecewise expressions, simplify separately on each side before comparing.

Solved Example: Factoring a removable form

Evaluate $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$.

Answer and Reasoning

Factor: $\frac{(x-5)(x+5)}{x-5} = x + 5$ for $x \neq 5$. Therefore the limit is 10.

Lesson Mastery Practice

21. Evaluate $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$.

Foundation

22. Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$.

Core

23. Evaluate $\lim_{x \rightarrow 2} \frac{x^4 - 16}{x - 2}$.

Advanced

24. Prove that for positive integer n , $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$.

Challenge

AP-Style Reasoning Check

$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ equals

(A) 0

(B) 3

(C) 6

(D) DNE

Lesson 1.7: Selecting Procedures for Determining Limits**Complete objectives**

- Diagnose a limit before calculating.
- Match algebraic structure to an efficient valid procedure.
- Choose among analytical, graphical, tabular, one-sided, and theorem-based evidence.
- Verify a result through a second representation or local check.

Strategic reminders

1. Use the cycle: diagnose → select → execute → verify.
2. Do not factor radicals, rationalize polynomials unnecessarily, or force algebra when only a graph/table is provided.
3. For composite and piecewise limits, track the direction of the inner landing value before selecting the outer branch.

Solved Example: Procedure choice

Choose the first move for $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$.

Answer and Reasoning

Substitution gives 0/0 and the numerator is a difference of squares, so factor and cancel.

Lesson Mastery Practice

25. Select the best first method for $\lim_{x \rightarrow 2} (3x^2 - 1)$.

Foundation

26. Substitution in a quotient gives 5/0. What should you inspect next?

Core

27. Choose a complete plan for $\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{x \sin(3x)}$.

Advanced

28. Design a decision tree with at least four branches for evaluating limits from formulas.

Challenge

AP-Style Reasoning Check

After substitution, $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$ has form $7/0$. The best next step is to

- | | |
|--------------|--|
| (A) cancel 7 | (B) inspect denominator signs from each side |
| (C) report 0 | (D) use the product law |

Lesson 1.8: Determining Limits Using the Squeeze Theorem**Complete objectives**

- State all hypotheses of the Squeeze Theorem.
- Create useful upper and lower bounds from absolute values and trigonometric inequalities.
- Distinguish persistent oscillation from shrinking oscillation.
- Write a complete AP-style squeeze justification.

Strategic reminders

1. Start from a known bound such as $-1 \leq \sin u \leq 1$ or $|\cos u| \leq 1$.
2. Multiply by a nonnegative shrinking factor or move to an absolute-value inequality.
3. End by naming the theorem and the common outer limit; merely saying “squeezed” is incomplete.

Solved Example: Shrinking oscillation

Evaluate $\lim_{x \rightarrow 0} x^2 \sin(1/x)$.

Answer and Reasoning

Because $|\sin(1/x)| \leq 1$, $|x^2 \sin(1/x)| \leq x^2 \rightarrow 0$. Therefore the limit is 0 by the Squeeze Theorem.

Lesson Mastery Practice

29. Given $-x^2 \leq f(x) \leq x^2$ near 0, find $\lim_{x \rightarrow 0} f(x)$.

Foundation

30. Evaluate $\lim_{x \rightarrow 0} |x|^{3/2} \sin(7/x)$.

Core

31. Evaluate $\lim_{x \rightarrow 0} \frac{x^3}{|x|} \sin(1/x)$.

Advanced

32. Prove: if u is bounded near c and $v(x) \rightarrow 0$, then $u(x)v(x) \rightarrow 0$.

Challenge

AP-Style Reasoning Check

If $-2|x - 1| \leq f(x) \leq 5|x - 1|$ near 1, then $\lim_{x \rightarrow 1} f(x)$ is

(A) -2

(B) 0

(C) 5

(D) not determined

Lesson 1.9: Connecting Multiple Representations of Limits**Complete objectives**

- Extract one-sided limits, two-sided limits, and point values from any representation.
- Translate accurately among formula, graph, table, words, and notation.
- Distinguish exact conclusions from numerical estimates.
- Audit whether multiple representations are locally coherent.

Strategic reminders

1. Create a fact ledger: left limit, right limit, two-sided conclusion, and point value.
2. When translating a formula into a graph, preserve excluded inputs and special assigned values.
3. For composite limits, include the direction from which the inner function reaches the outer target.

Solved Example: Formula to graph facts

Let $f(x) = \frac{x^2-1}{x-1}$ for $x \neq 1$ and $f(1) = 6$. Describe the graph and state the limit.

Answer and Reasoning

For nearby $x \neq 1$, $f(x) = x + 1$, so the graph is the line $y = x + 1$ with an open circle at $(1, 2)$ and a filled point at $(1, 6)$. The limit is 2.

Lesson Mastery Practice

- 33.** A graph approaches 3 from both sides at $x = 2$ and has a filled point at 7. Write the limit and point value.

Foundation

- 34.** Use Representation Set R to state the left-hand limit, right-hand limit, and two-sided limit at 0.

Core

- 35.** A graph shows a hole at height 5, but a table rounded to the nearest integer shows 4 on the left and 6 on the right. Explain how both could describe the same limit.

Advanced

36. Create five coherent representations for a function with $\lim_{x \rightarrow 1} f(x) = 4$ and $f(1) = -3$: notation, words, formula, graph description, and table description.

Challenge

AP-Style Reasoning Check

Which pair of statements is coherent?

- (A) $\lim_{x \rightarrow 2} f(x) = 3$ and a graph approaching 5 from both sides
- (B) $f(2) = 7$ and a filled point at $(2, 7)$
- (C) $\lim_{x \rightarrow 2^-} f(x) = 1$ and a left table approaching 4
- (D) A formula equal to $x + 1$ near 2 and a hole at height 8

Lesson 1.10: Exploring Types of Discontinuities**Complete objectives**

- Classify removable, jump, and infinite discontinuities using one-sided evidence.
- Distinguish the three named types from oscillatory or domain-gap failures.
- Use factor cancellation and multiplicity to classify rational-function discontinuities.
- Write the limit evidence before naming the type.

Strategic reminders

1. For rational functions, factor first: complete cancellation creates a hole; a remaining denominator factor creates a pole.
2. For piecewise functions, evaluate the branch limits at every breakpoint.
3. Do not call every DNE limit a jump; identify the actual one-sided fingerprint.

Solved Example: Removable rational discontinuity

Classify $f(x) = \frac{x^2-9}{x-3}$ at $x = 3$.

Answer and Reasoning

For $x \neq 3$, $f(x) = x + 3$, so the common finite limit is 6 while the original function is undefined. The discontinuity is removable.

Lesson Mastery Practice

37. Classify $\frac{x^2-16}{x-4}$ at $x = 4$.

Foundation

38. Classify every discontinuity of $f(x) = \frac{(x-3)(x+2)}{(x-3)(x-1)}$.

Core

39. Classify all discontinuities of $\frac{x^2-1}{(x-1)^2(x+3)}$.

Advanced

40. Construct one function having a removable discontinuity at -1 , a jump at 2 , and an infinite discontinuity at 4 .

Challenge

AP-Style Reasoning Check

Which evidence identifies a jump discontinuity at $x = c$?

- | | |
|--|--|
| (A) Both sides approach the same finite value. | (B) Finite one-sided limits exist but are unequal. |
| (C) Both sides approach $+\infty$. | (D) The function value is missing. |

Lesson 1.11: Defining Continuity at a Point**Complete objectives**

- State and apply the three conditions for continuity at a point.
- Identify the first failed condition from graphs, tables, formulas, and piecewise rules.
- Use continuity of standard functions and valid compositions.
- Reject common invalid converses about limits and point values.

Strategic reminders

1. Audit in order: domain, two-sided limit, equality.
2. For a formula built from standard continuous functions, check the domain and then substitute.
3. A corner may be continuous; differentiability is not required for continuity.

Solved Example: All conditions hold

Verify continuity of $f(x) = \sqrt{x+5}$ at $x = 4$.

Answer and Reasoning

$f(4) = 3$ exists. The square-root function is continuous on its domain, so $\lim_{x \rightarrow 4} f(x) = 3 = f(4)$. All three conditions hold.

Lesson Mastery Practice

41. List the three continuity conditions at $x = c$.

Foundation

42. For $f(x) = |x - 1|$, verify continuity at $x = 1$.

Core

43. Determine whether $f(x) = \frac{x^2-9}{x-3}$ is continuous at 3 as originally defined.

Advanced

44. Prove that if f is continuous at c and $f(c) > 0$, then $f(x) > 0$ for all x sufficiently close to c .

Challenge

AP-Style Reasoning Check

Which condition is not required for continuity at c ?

(A) $f(c)$ exists

(B) $\lim_{x \rightarrow c} f(x)$ exists

(C) $\lim_{x \rightarrow c} f(x) = f(c)$

(D) $f'(c)$ exists

Lesson 1.12: Confirming Continuity over an Interval**Complete objectives**

- Distinguish continuity on open, closed, and half-open intervals.
- Use one-sided continuity at included endpoints.
- Find maximal intervals of continuity from domain restrictions.
- Justify continuity of sums, products, quotients, roots, and compositions on stated intervals.

Strategic reminders

1. Find the real domain before writing intervals.
2. Split at denominator zeros, logarithm boundaries, even-root restrictions, and piecewise joins.
3. Use square brackets only when the endpoint is in the domain and the correct one-sided continuity condition holds.

Solved Example: Rational intervals

Find the maximal intervals of continuity of $f(x) = \frac{x+1}{x^2-4}$.

Answer and Reasoning

The denominator vanishes at ± 2 . The maximal intervals are $(-\infty, -2)$, $(-2, 2)$, and $(2, \infty)$.

Lesson Mastery Practice

45. Find the intervals of continuity of $f(x) = 1/(x - 5)$.

Foundation

46. Find maximal intervals for $\frac{x^2+1}{(x-1)(x+4)}$.

Core

47. Find maximal intervals for $f(x) = \frac{\sqrt{x+1}}{x-3}$.

Advanced

48. Prove that $\max\{f, g\}$ is continuous wherever f and g are continuous.

Challenge

AP-Style Reasoning Check

The maximal intervals of continuity of $1/(x^2 - 9)$ are

- (A) $(-\infty, \infty)$ (B) $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$ as three intervals
(C) $[-3, 3]$ (D) $(0, \infty)$

Lesson 1.13: Removing Discontinuities**Complete objectives**

- Decide whether a discontinuity can be removed by changing a point value.
- Find parameters that make piecewise functions continuous at joins.
- Create or cancel common factors to identify removable holes.
- Solve multi-boundary continuity systems and justify uniqueness or impossibility.

Strategic reminders

1. First classify the discontinuity. A point-value repair is possible only when a finite common limit already exists.
2. For piecewise joins, write equations before solving parameters.
3. For rational expressions, use the factor theorem: $x - c$ divides a polynomial $P(x)$ exactly when $P(c) = 0$.

Solved Example: Fill a hole

Define $f(3)$ so that $f(x) = \frac{x^2-9}{x-3}$ is continuous at 3.

Answer and Reasoning

For $x \neq 3$, $f(x) = x + 3$, so the limit is 6. Define $f(3) = 6$.

Lesson Mastery Practice

49. What value repairs $f(x) = \frac{x^2-4}{x-2}$ at $x = 2$?

Foundation

50. Find k so $f(x) = \begin{cases} 2x + k, & x < 1 \\ x^2 + 3, & x \geq 1 \end{cases}$ is continuous at 1.

Core

51. Find a, b so $f(x) = \begin{cases} ax + b, & x < 1 \\ x^2, & 1 \leq x < 3 \\ 2x + b, & x \geq 3 \end{cases}$ is continuous at both joins.

Advanced

52. For positive integer n , choose coefficients so $\frac{x^n - a^n}{x - a}$ extends continuously to $x = a$, and give the extension value.

Challenge

AP-Style Reasoning Check

To repair a removable discontinuity at $x = c$, define $f(c)$ to be

- | | |
|--------------------------------|------------------------------|
| (A) 0 | (B) the left-hand limit only |
| (C) the finite two-sided limit | (D) any real number |

Lesson 1.14: Connecting Infinite Limits and Vertical Asymptotes**Complete objectives**

- Interpret positive and negative infinite limits as directional unbounded behavior.
- Determine one-sided signs near denominator zeros.
- Use factor multiplicity and cancellation to distinguish holes from vertical asymptotes.
- Connect formulas, sign charts, tables, graphs, and asymptote notation.

Strategic reminders

1. Factor and cancel before building the sign chart.
2. For each candidate denominator zero, choose a test point just left and just right.
3. State both one-sided limits; merely listing the asymptote line is incomplete when signs are requested.

Solved Example: Odd pole

Analyze $f(x) = \frac{2}{x-3}$ near $x = 3$.

Answer and Reasoning

The numerator is positive. The denominator is negative on the left and positive on the right, so the limits are $-\infty$ and $+\infty$.

Lesson Mastery Practice

53. Find the vertical asymptote of $1/(x - 6)$.

Foundation

54. Analyze $\frac{-3}{x+4}$ at $x = -4$.

Core

55. Find all vertical asymptotes and one-sided signs of $\frac{x+2}{(x-1)(x+3)}$.

Advanced

56. Construct a rational function with a hole at 1, an even pole at -2 opening downward, and an odd pole at 4 with left $+\infty$.

Challenge

AP-Style Reasoning Check

For $f(x) = 1/(x - 2)^2$, the one-sided limits at 2 are

- | | |
|--------------------------------------|--------------------|
| (A) both $-\infty$ | (B) both $+\infty$ |
| (C) left $-\infty$, right $+\infty$ | (D) finite |

Lesson 1.15: Connecting Limits at Infinity and Horizontal Asymptotes**Complete objectives**

- Evaluate limits as $x \rightarrow \infty$ and $x \rightarrow -\infty$.
- Determine horizontal asymptotes from end limits.
- Use degree comparison, dominant powers, radicals, conjugates, and growth hierarchy.
- Analyze the two ends independently and handle $\sqrt{x^2} = |x|$ correctly.

Strategic reminders

1. For rational functions divide by the highest power in the denominator or compare leading terms.
2. For radical differences, rationalize before comparing dominant terms.
3. A curve may cross a horizontal asymptote; an asymptote describes end behavior, not a barrier.

Solved Example: Equal degrees

Evaluate $\lim_{x \rightarrow \pm\infty} \frac{3x^2 - x}{2x^2 + 5}$.

Answer and Reasoning

Dividing by x^2 gives a limit of $3/2$ at both ends, so $y = 3/2$ is a horizontal asymptote.

Lesson Mastery Practice

57. Evaluate $\lim_{x \rightarrow \infty} \frac{3x+1}{x-4}$.

Foundation

58. Evaluate $\lim_{x \rightarrow \pm\infty} \frac{5x^3 - x}{2x^3 + 1}$.

Core

59. Evaluate $\lim_{x \rightarrow \pm\infty} \frac{2x^2 + 1}{\sqrt{4x^4 + 3}}$.

Advanced

60. Construct a function with right horizontal asymptote $y = 2$ and left horizontal asymptote $y = -3$.

Challenge

AP-Style Reasoning Check

$\lim_{x \rightarrow \infty} \frac{4x^2+1}{2x^2-3x}$ equals

(A) 0

(B) 2

(C) 4

(D) $+\infty$

Lesson 1.16: Working with the Intermediate Value Theorem**Complete objectives**

- State every hypothesis of the Intermediate Value Theorem.
- Use endpoint values to guarantee roots or other target outputs.
- Count minimum distinct solutions from disjoint sign-changing intervals.
- Use auxiliary functions and bisection to localize guaranteed solutions.

Strategic reminders

1. State continuity, endpoint values, the bracketing inequality, and the existence conclusion.
2. Use disjoint intervals to guarantee distinct solutions.
3. For bisection, retain the half-interval whose endpoint values still bracket the target.

Solved Example: Guarantee a root

Show $x^3 + x - 1 = 0$ has a solution in $(0, 1)$.

Answer and Reasoning

Let $f(x) = x^3 + x - 1$, continuous on $[0, 1]$. Since $f(0) = -1$ and $f(1) = 1$, 0 lies between the endpoint values. IVT guarantees $c \in (0, 1)$ with $f(c) = 0$.

Lesson Mastery Practice

- 61.** List the three main IVT hypotheses/conclusion components.

Foundation

- 62.** Show $x^2 - 3 = 0$ has a solution in $(1, 2)$.

Core

- 63.** Show $\cos x = x$ has a solution in $(0, 1)$.

Advanced

64. Prove that any odd-degree real polynomial has at least one real root.

Challenge

AP-Style Reasoning Check

Which condition is required by IVT?

(A) Differentiability on (a, b)

(B) Continuity on $[a, b]$

(C) $f(a) = f(b)$

(D) A unique root

Unit 1 Difficult and Challenge Synthesis

1. Using the precise definition, prove that $\lim_{x \rightarrow 2} x^2 = 4$. Give an explicit $\delta(\varepsilon)$.

Difficult

2. Write the logical negation of $\lim_{x \rightarrow c} f(x) = L$, then use it to prove that $\lim_{x \rightarrow 0} \operatorname{sgn}(x)$ does not exist.

Challenge

3. Let $g(x) \rightarrow 2^-$ as $x \rightarrow 1$, and let $f(t) = t^2$ for $t < 2$ and $f(t) = 7 - t$ for $t \geq 2$. Find $\lim_{x \rightarrow 1} f(g(x))$ and explain why the direction matters.

Hard

4. Find constants a, b such that $f(x) = \frac{x^2 + ax + b}{x^2 - x - 2}$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = -1$. Then find the hole coordinates.

Difficult

5. Evaluate $\lim_{x \rightarrow 0} x^2 \sin(1/x^3) \cos(e^{1/x^4})$ and give a complete squeeze argument.

Hard

6. Analyze every discontinuity and one-sided infinite limit of $r(x) = \frac{(x-3)^2(x+1)}{(x-3)(x+2)^2(x-1)}$.

Challenge

7. Evaluate both $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 3x} - x)$ and $\lim_{x \rightarrow -\infty} (\sqrt{x^2 + 3x} + x)$.

Challenge

8. A continuous function satisfies $f(0) = -3$, $f(1) = 2$, $f(2) = -1$, and $f(4) = 5$. What is the greatest minimum number of distinct zeros that is guaranteed? Justify.

Hard

Cumulative AP-Style Checkpoint

1. Which statement is the precise meaning of $\lim_{x \rightarrow c} f(x) = L$?

- (A) There exists $\varepsilon > 0$ such that every $\delta > 0$ works
 (B) $f(c) = L$ and f is differentiable at c
 (C) For every $\delta > 0$ there exists $\varepsilon > 0$ depending on x
 (D) For every $\varepsilon > 0$ there exists $\delta > 0$ such that $0 < |x - c| < \delta$ implies $|f(x) - L| < \varepsilon$

2. To prove $\lim_{x \rightarrow 2} x^2 = 4$, which choice of δ is sufficient for every $\varepsilon > 0$?

- (A) 5ε
 (B) $\min\{5, 1/\varepsilon\}$
 (C) $\min\{1, \varepsilon/5\}$
 (D) $\varepsilon + 5$

3. For $f(x) = x^2 - 4x$, which expression represents the instantaneous rate at $x = 3$?

- (A) $\frac{f(3) - f(3)}{3 - 3}$
 (B) $\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$
 (C) $\lim_{x \rightarrow 0} \frac{f(x) - f(3)}{x - 3}$
 (D) $\frac{f(4) - f(2)}{2}$

4. A particle has $s(t) = 5t^2 - t$. Its instantaneous velocity at $t = 2$ is

- (A) 9
 (B) 10
 (C) 19
 (D) 20

5. For $P(t) = \frac{800}{1 + 3e^{-0.4t}}$, a calculator gives $P(4.99) \approx 568.330$ and $P(5.01) \approx 569.645$. Which is the best estimate of $P'(5)$?

- (A) 0.658
 (B) 6.58
 (C) 65.8
 (D) 131.5

6. If $\lim_{x \rightarrow 2^-} f(x) = 5$ and $\lim_{x \rightarrow 2^+} f(x) = 5$, which must be true?

- (A) $f(2) = 5$
 (B) $f(2)$ exists
 (C) $\lim_{x \rightarrow 2} f(x) = 5$
 (D) f is differentiable at 2

7. Which notation means that $g(x)$ decreases without bound as x approaches 4 through values less than 4?

- (A) $\lim_{x \rightarrow 4^-} g(x) = -\infty$
 (B) $\lim_{x \rightarrow 4^+} g(x) = -\infty$
 (C) $\lim_{x \rightarrow -4^-} g(x) = +\infty$
 (D) $g(4) = -\infty$

8. A calculator table produces the data below. Which notation best describes the behavior?

x	1.99	1.999	2.001	2.01
$g(x)$	-48.7	-498.7	501.3	51.3

- (A) $\lim_{x \rightarrow 2} g(x) = 0$
 (B) $\lim_{x \rightarrow 2^-} g(x) = -\infty$ and $\lim_{x \rightarrow 2^+} g(x) = +\infty$
 (C) $\lim_{x \rightarrow 2} g(x) = +\infty$
 (D) $g(2) = 0$

9. At $x = a$, a graph approaches 6 from both sides but has a filled point at height -3 . Which statement is true?

- (A) $\lim_{x \rightarrow a} f(x) = -3$
 (B) $\lim_{x \rightarrow a} f(x) = 6$
 (C) The limit DNE.
 (D) $f(a) = 6$

10. Which graphical feature necessarily prevents a finite two-sided limit at the target?

- (A) A corner
 (B) A cusp whose sides meet
 (C) A jump with unequal one-sided heights
 (D) A hole

11. A calculator graph suggests a limit of 1.732 at $x = 2$, but the curve is steep and the window uses y-scale 10. What is the best next step?
- (A) Report 1.732000 immediately. (B) Zoom and confirm with a numerical table from both sides.
- (C) Read the filled point only. (D) Conclude DNE.
12. A table near $x = 5$ has outputs approaching -2 from both sides, while the center row is 17. Which conclusion is best?
- (A) $\lim_{x \rightarrow 5} f(x) = 17$ (B) $\lim_{x \rightarrow 5} f(x) = -2$
- (C) The limit DNE. (D) No conclusion is possible.
13. Which input list is best for estimating $\lim_{x \rightarrow 3} f(x)$?
- (A) 0, 1, 2, 3 (B) 2.9, 2.99, 3.01, 3.1
- (C) 3, 4, 5, 6 (D) $-3, -2, -1, 0$
14. A calculator table for $h(x) = \frac{e^x - 1}{x}$ gives 0.99950, 0.99995, 1.00005, 1.00050 near $x = 0$. The best estimate is
- (A) 0 (B) 0.5
- (C) 1 (D) DNE
15. If $\lim f = 2$ and $\lim g = -5$, then $\lim(3f - g^2)$ equals
- (A) -19 (B) -7
- (C) 19 (D) 31
16. Which condition is required to use the quotient law for $\lim f(x)/g(x)$?
- (A) $f(c)$ exists (B) $g(c) \neq 0$
- (C) $\lim g(x) \neq 0$ (D) f and g are differentiable
17. A calculator gives $u(x) \rightarrow 1.73205$ and $v(x) \rightarrow -0.50000$. Estimate $\lim[u(x)^2 + 4v(x)]$.
- (A) -1 (B) 0
- (C) 1 (D) 5
18. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ equals
- (A) 0 (B) 3
- (C) 6 (D) DNE
19. $\lim_{x \rightarrow 0} \frac{\sin(4x)}{3x}$ equals
- (A) $\frac{3}{4}$ (B) $\frac{4}{3}$
- (C) 3 (D) 4
20. A calculator suggests $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} \approx 2$. Which table best supports the conclusion?
- (A) Only $x = 0$ (B) Inputs near 0 from both sides with increasing precision
- (C) Only positive integers (D) Only $x = 1, 2, 3$

Cumulative Free-Response Synthesis

FRQ1 – Limits from numerical and graphical evidence

A function is sampled near $x = 2$ with

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$	3.61	3.9601	3.996001	4.004001	4.0401	4.41

(a) Estimate $\lim_{x \rightarrow 2} f(x)$ and justify from both sides. [2 points]

(b) Give a simple formula consistent with the table and state whether $f(2)$ is determined by the data. [2 points]

(c) Explain one way rounding could create a misleading conclusion. [2 points]

FRQ2 – Algebraic limit and continuity repair

Let $g(x) = \frac{x^2 - 5x + 6}{x - 2}$ for $x \neq 2$.

(a) Evaluate $\lim_{x \rightarrow 2} g(x)$ and identify the algebraic procedure. [3 points]

(b) Define $g(2)$ to make the extended function continuous. [1 points]

(c) Explain why cancellation does not prove the original function was defined at 2. [2 points]

FRQ3 – Precise definition of a limit

Prove from the ε - δ definition that $\lim_{x \rightarrow 2} x^2 = 4$.

(a) Explain why imposing $|x - 2| < 1$ produces a useful bound on $|x + 2|$. [2 points]

(b) Choose an explicit $\delta(\varepsilon)$ and complete the proof. [4 points]

(c) State why the condition $0 < |x - 2|$ is present. [1 points]

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